

Geometric interpretation of pd envelope

Relative deRham stack

Let X be a closed subscheme of Y , over a base scheme S . For simplicity, we use X^{dR} to denote the relative derham stack over S . In char 0, we have the Cartersian diagram

$$\begin{array}{ccc} Y_X^f & \longrightarrow & X^{dR} \\ \downarrow & \lrcorner & \downarrow \\ Y & \longrightarrow & Y^{dR} \end{array}$$

where Y_X^f is the formal completion of Y along X .

In char p , if $X \hookrightarrow Y$ is a closed immersion of smooth schemes over S , we have the Cartesian diagram

$$\begin{array}{ccc} Y_X^\gamma & \longrightarrow & X^{dR} \\ \downarrow & \lrcorner & \downarrow \\ Y & \longrightarrow & Y^{dR} \end{array}$$

where Y_X^γ is the pd envelope of the closed immersion $X \hookrightarrow Y$. If we are derived enough, the smoothness condition is superfluous.

Slogan: pd envelope is the relative deRham stack.

This explains why there is a Y -connection on Y_X^γ .

For a ring A and $a \in A$, a pd structure on a is a homomorphism $\mathbb{Z}[\frac{x^n}{n!}] \rightarrow A$ sending x to a .

The pd envelope of $\mathbb{Z}[x_1, \dots, x_m, y_1, \dots, y_n] \rightarrow \mathbb{Z}[x_1, \dots, x_m]$ is $\mathbb{Z}[x_1, \dots, x_m, \frac{y_1^r}{r!}, \dots, \frac{y_n^r}{r!}]$. Every (animated) ring surjection $A \rightarrow A/I$ is a sifted colimit of this type, and the (derived) pd envelope is the correspoding sifted colimit of pd envelope. If I is generated by regular sequence, then the derived construction becomes classical.

The morphism $Y_X^f \rightarrow Y$ is faithfully flat. In char p , the morphism $Y_X^\gamma \rightarrow Y$ factors through the Frobenius neighborhood Y_X^ϕ of X in Y . Recall the Frobenius neighborhood is defined by the Cartesian diagram

$$\begin{array}{ccc} Y_X^\phi & \hookrightarrow & Y \\ \phi_{X/Y} \downarrow & \lrcorner & \downarrow F_Y \\ X & \hookrightarrow & Y \end{array}$$

The Frobenius neighborhood of A/I in A is $A/I^{(p)}$, where $I^{(p)} := \{f^p \mid f \in I\}$. Notice that if I is generated by regular sequence, then so is $I^{(p)}$. As the natural map $A \rightarrow A_I^\gamma$

kills $I^{(p)}$, we see there is indeed a factorization $Y_X^\gamma \rightarrow Y_X^\phi$. We can also get this map via transmutation.

$$\begin{array}{ccccc}
 & & Y_X^\phi & \xrightarrow{\quad} & Y \\
 & \nearrow & \downarrow & & \downarrow \\
 Y_X^\gamma & \xrightarrow{\quad} & Y & \xrightarrow{\quad} & Y \\
 \downarrow & & \downarrow & & \downarrow \\
 & \nearrow & X^{(1)} & \xrightarrow{\quad} & Y^{(1)} \\
 X^{dR} & \xrightarrow{\quad} & Y^{dR} & &
 \end{array}$$

p-curvature

Suppose Y is smooth over S , and consider the closed immersion $Y \rightarrow Y(1) := Y \times_S Y$. We use $Y^\phi(1), Y^\gamma(1)$ to denote corresponding Frobenius neighborhood and pd envelope. Notice $Y^\phi(1), Y^\gamma(1)$ are both groupoids over Y , with $[Y/Y^\phi(1)] = Y^{(1)}$ and $[Y/Y^\gamma(1)] = Y^{dR}$.

Let A be a groupoid over Y . Consider the loop groupoid L defined by

$$\begin{array}{ccc}
 L & \xrightarrow{\quad} & A \\
 \downarrow & \lrcorner & \downarrow (s,t) \\
 Y & \xrightarrow{\Delta} & Y \times_S Y
 \end{array}$$

Then clearly L is a group scheme over Y , and further a group scheme over $Y \times_S Y$ since the diagonal is a monomorphism. Then the stack $[A/L]$ is still a groupoid over Y . In fact, from the diagram

$$\begin{array}{ccccc}
 L & \xrightarrow{\quad} & Y \times_F Y & \xrightarrow{\quad} & F \\
 \downarrow & \lrcorner & \downarrow (s,t) & \lrcorner & \downarrow \\
 Y & \xrightarrow{\Delta} & Y \times_S Y & \xrightarrow{\quad} & F \times_S F
 \end{array}$$

we see that L is actually the loop stack of $F \times_S Y$, where $F = [Y/A]$.

Remark 1.1. How to think of $[A/L]$ in terms of stacks?

We will be in the good case that $[A/L]$ is actually a scheme. Namely, we take $A = Y^\gamma(1)$. It can be shown that $L = T_{Y/S}^{(p),\gamma}$ and $[A/L] = Y^\phi(1)$.

Vector bundles, or quasicoherent sheaves on Y with actions of A are equivalent modules on Y with integrable connection and quasi-nilpotent p -curvature. The p -curvature is exactly given by the action of L .